

Application of the Sieve Streaming Algorithm

Question 1: Select a Subset of Size $k = 3$

Initialization:

- **Threshold Set** $O = \{4 \times i \mid i \in \mathbb{Z}^+\}$ (i.e., $O = \{4, 8, 12, 16, \dots\}$)
- **Desired Subset Size:** $k = 3$
- **Initialize:** $m = 0$
- **Initialize:** $S_v = \emptyset$ for necessary v (maintained lazily).

Processing the Stream: The items arrive from right to left:

$$e_1 = \{1\}, e_2 = \{2, 3\}, e_3 = \{3, 7\}, e_4 = \{2, 4, 5\}, e_5 = \{1, 2, 3\}, e_6 = \{1, 5, 6, 8\}$$

At each step:

- **Update** m : $m = \max(m, f(\{e_i\}))$
- **Define** $O_i = \{4 \times i \mid m \leq 4i \leq 2km\}$
- **Delete** S_v where $v \notin O_i$
- For each $v \in O_i$:
 - If $\Delta f(e_i \mid S_v) \geq \frac{v/2 - f(S_v)}{k - |S_v|}$ and $|S_v| < k$, then add e_i to S_v .

Detailed Execution:

Step 1: Processing $e_1 = \{1\}$

- Update m : $m = \max(0, 1) = 1$
- Compute O_1 : $O_1 = \{4\}$ (since $m = 1$ and $2km = 6$)
- Process e_1 for $v = 4$:
 - **Current** $S_4 = \emptyset$, $f(S_4) = 0$, $|S_4| = 0$, $k - |S_4| = 3$
 - Marginal Gain: $\Delta f(e_1 \mid S_4) = 1 - 0 = 1$
 - Compute RHS: $\frac{v/2 - f(S_4)}{k - |S_4|} = \frac{2 - 0}{3} = \frac{2}{3} \approx 0.6667$
 - Condition: $1 \geq 0.6667$ (True)
 - Action: Add e_1 to S_4 : $S_4 = \{\{1\}\}$

Step 2: Processing $e_2 = \{2, 3\}$

- Update m : $m = \max(1, 2) = 2$
- Compute O_2 : $O_2 = \{4, 8, 12\}$ (since $m = 2$ and $2km = 12$)

- Process e_2 for each $v \in O_2$:
 - For $v = 4$:
 - * **Current** $S_4 = \{\{1\}\}$, $f(S_4) = 1$, $|S_4| = 1$, $k - |S_4| = 2$
 - * Marginal Gain: $\Delta f(e_2 \mid S_4) = 3 - 1 = 2$
 - * Compute RHS: $\frac{2-1}{2} = 0.5$
 - * Condition: $2 \geq 0.5$ (True)
 - * Action: Add e_2 to S_4 : $S_4 = \{\{1\}, \{2, 3\}\}$
 - For $v = 8$:
 - * **Initialize** $S_8 = \emptyset$
 - * Marginal Gain: $\Delta f(e_2 \mid S_8) = 2 - 0 = 2$
 - * Compute RHS: $\frac{4-0}{3} \approx 1.3333$
 - * Condition: $2 \geq 1.3333$ (True)
 - * Action: Add e_2 to S_8 : $S_8 = \{\{2, 3\}\}$
 - For $v = 12$:
 - * **Initialize** $S_{12} = \emptyset$
 - * Marginal Gain: $\Delta f(e_2 \mid S_{12}) = 2$
 - * Compute RHS: $\frac{6-0}{3} = 2$
 - * Condition: $2 \geq 2$ (True)
 - * Action: Add e_2 to S_{12} : $S_{12} = \{\{2, 3\}\}$

Step 3: Processing $e_3 = \{3, 7\}$

- Update m : $m = 2$ (unchanged)
- Compute $O_3 = O_2 = \{4, 8, 12\}$
- Process e_3 for each $v \in O_3$:
 - For $v = 4$:
 - * **Current** $S_4 = \{\{1\}, \{2, 3\}\}$, $f(S_4) = 3$, $|S_4| = 2$, $k - |S_4| = 1$
 - * Marginal Gain: $\Delta f(e_3 \mid S_4) = 4 - 3 = 1$
 - * Compute RHS: $\frac{2-3}{1} = -1$
 - * Condition: $1 \geq -1$ (True)
 - * Action: Add e_3 to S_4 : $S_4 = \{\{1\}, \{2, 3\}, \{3, 7\}\}$
 - For $v = 8$:
 - * **Current** $S_8 = \{\{2, 3\}\}$, $f(S_8) = 2$, $|S_8| = 1$, $k - |S_8| = 2$
 - * Marginal Gain: $\Delta f(e_3 \mid S_8) = 3 - 2 = 1$
 - * Compute RHS: $\frac{4-2}{2} = 1$
 - * Condition: $1 \geq 1$ (True)
 - * Action: Add e_3 to S_8 : $S_8 = \{\{2, 3\}, \{3, 7\}\}$
 - For $v = 12$:

- * **Current** $S_{12} = \{\{2, 3\}\}$, $f(S_{12}) = 2$, $|S_{12}| = 1$
- * Marginal Gain: $\Delta f(e_3 \mid S_{12}) = 3 - 2 = 1$
- * Compute RHS: $\frac{6-2}{2} = 2$
- * Condition: $1 \geq 2$ (False)
- * Action: Do not add e_3 to S_{12}

Step 4: Processing $e_4 = \{2, 4, 5\}$

- Update m : $m = \max(2, 3) = 3$
- Compute $O_4 = \{4, 8, 12, 16\}$ (since $m = 3$ and $2km = 18$)
- Process e_4 for each $v \in O_4$:
 - For $v = 4$:
 - * S_4 is full ($|S_4| = k$)
 - * Action: Do not add e_4 to S_4
 - For $v = 8$:
 - * **Current** $S_8 = \{\{2, 3\}, \{3, 7\}\}$, $f(S_8) = 3$, $|S_8| = 2$, $k - |S_8| = 1$
 - * Marginal Gain: $\Delta f(e_4 \mid S_8) = 5 - 3 = 2$
 - * Compute RHS: $\frac{4-3}{1} = 1$
 - * Condition: $2 \geq 1$ (True)
 - * Action: Add e_4 to S_8 : $S_8 = \{\{2, 3\}, \{3, 7\}, \{2, 4, 5\}\}$
 - For $v = 12$:
 - * **Current** $S_{12} = \{\{2, 3\}\}$, $f(S_{12}) = 2$, $|S_{12}| = 1$
 - * Marginal Gain: $\Delta f(e_4 \mid S_{12}) = 4 - 2 = 2$
 - * Compute RHS: $\frac{6-2}{2} = 2$
 - * Condition: $2 \geq 2$ (True)
 - * Action: Add e_4 to S_{12} : $S_{12} = \{\{2, 3\}, \{2, 4, 5\}\}$
 - For $v = 16$:
 - * **Initialize** $S_{16} = \emptyset$
 - * Marginal Gain: $\Delta f(e_4 \mid S_{16}) = 3$
 - * Compute RHS: $\frac{8-0}{3} \approx 2.6667$
 - * Condition: $3 \geq 2.6667$ (True)
 - * Action: Add e_4 to S_{16} : $S_{16} = \{\{2, 4, 5\}\}$

Step 5: Processing $e_5 = \{1, 2, 3\}$

- Update m : $m = 3$ (unchanged)
- Compute $O_5 = O_4 = \{4, 8, 12, 16\}$
- Process e_5 for each $v \in O_5$:

- For $v = 4$:
 - * S_4 is full ($|S_4| = k$)
 - * Action: Do not add e_5 to S_4
- For $v = 8$:
 - * S_8 is full ($|S_8| = k$)
 - * Action: Do not add e_5 to S_8
- For $v = 12$:
 - * **Current** $S_{12} = \{\{2, 3\}, \{2, 4, 5\}\}$, $f(S_{12}) = 4$, $|S_{12}| = 2$
 - * Marginal Gain: $\Delta f(e_5 \mid S_{12}) = 5 - 4 = 1$
 - * Compute RHS: $\frac{6-4}{1} = 2$
 - * Condition: $1 \geq 2$ (False)
 - * Action: Do not add e_5 to S_{12}
- For $v = 16$:
 - * **Current** $S_{16} = \{\{2, 4, 5\}\}$, $f(S_{16}) = 3$, $|S_{16}| = 1$
 - * Marginal Gain: $\Delta f(e_5 \mid S_{16}) = 5 - 3 = 2$
 - * Compute RHS: $\frac{8-3}{2} = 2.5$
 - * Condition: $2 \geq 2.5$ (False)
 - * Action: Do not add e_5 to S_{16}

Step 6: Processing $e_6 = \{1, 5, 6, 8\}$

- Update m : $m = \max(3, 4) = 4$
- Compute $O_6 = \{4, 8, 12, 16, 20, 24\}$ (since $m = 4$ and $2km = 24$)
- Process e_6 for each $v \in O_6$:
 - For $v = 4$:
 - * S_4 is full ($|S_4| = k$)
 - * Action: Do not add e_6 to S_4
 - For $v = 8$:
 - * S_8 is full ($|S_8| = k$)
 - * Action: Do not add e_6 to S_8
 - For $v = 12$:
 - * **Current** $S_{12} = \{\{2, 3\}, \{2, 4, 5\}\}$, $f(S_{12}) = 4$, $|S_{12}| = 2$
 - * Marginal Gain: $\Delta f(e_6 \mid S_{12}) = 7 - 4 = 3$
 - * Compute RHS: $\frac{6-4}{1} = 2$
 - * Condition: $3 \geq 2$ (True)
 - * Action: Add e_6 to S_{12} : $S_{12} = \{\{2, 3\}, \{2, 4, 5\}, \{1, 5, 6, 8\}\}$
 - For $v = 16$:
 - * **Current** $S_{16} = \{\{2, 4, 5\}\}$, $f(S_{16}) = 3$, $|S_{16}| = 1$

- * Marginal Gain: $\Delta f(e_6 \mid S_{16}) = 6 - 3 = 3$
- * Compute RHS: $\frac{8-3}{2} = 2.5$
- * Condition: $3 \geq 2.5$ (True)
- * Action: Add e_6 to S_{16} : $S_{16} = \{\{2, 4, 5\}, \{1, 5, 6, 8\}\}$
- For $v = 20$ and $v = 24$:
 - * **Initialize** $S_v = \emptyset$
 - * Marginal Gain: $\Delta f(e_6 \mid S_v) = 4$
 - * Compute RHS for $v = 20$: $\frac{10-0}{3} \approx 3.3333$
 - * Condition: $4 \geq 3.3333$ (True)
 - * Action: Add e_6 to S_{20} : $S_{20} = \{\{1, 5, 6, 8\}\}$
 - * Compute RHS for $v = 24$: $\frac{12-0}{3} = 4$
 - * Condition: $4 \geq 4$ (True)
 - * Action: Add e_6 to S_{24} : $S_{24} = \{\{1, 5, 6, 8\}\}$

Final Selected Subset

Select S_{12} as it has the highest $f(S_v)$ among all S_v : $S_{12} = \{\{2, 3\}, \{2, 4, 5\}, \{1, 5, 6, 8\}\}$.

- **Total Unique Elements Covered:** $\bigcup_{S_i \in S_{12}} S_i = \{1, 2, 3, 4, 5, 6, 8\}$
- **Function Value:** $f(S_{12}) = 7$

Answer to Question 1

We select the subset:

$$S = \{\{2, 3\}, \{2, 4, 5\}, \{1, 5, 6, 8\}\}$$

This subset covers 7 unique elements: $\{1, 2, 3, 4, 5, 6, 8\}$.

Question 2: Number of Times m was Updated

We updated m a total of 4 times:

- After e_1 : m updated from 0 to 1
- After e_2 : m updated from 1 to 2
- After e_4 : m updated from 2 to 3
- After e_6 : m updated from 3 to 4

Question 3: Optimal Solution and Fraction of Optimal Solution

- **Optimal Solution Subsets:** $\{\{1, 5, 6, 8\}, \{2, 4, 5\}, \{3, 7\}\}$
- **Unique Elements Covered:** $\{1, 2, 3, 4, 5, 6, 7, 8\}$
- **Optimal Value:** $f(\text{Optimal}) = 8$
- **Streaming Algorithm Value:** $f(S) = 7$
- **Fraction of Optimal Value Obtained:**

$$\text{Fraction} = \frac{f(S)}{f(\text{Optimal})} = \frac{7}{8} = 0.875$$

Summary:

- **Selected Subset (Streaming Algorithm):** $S = \{\{2, 3\}, \{2, 4, 5\}, \{1, 5, 6, 8\}\}$
- **Function Value:** $f(S) = 7$
- **Optimal Function Value:** $f(\text{Optimal}) = 8$
- **Fraction of Optimal Value:** 87.5%

This completes the detailed application of the sieve streaming algorithm to select a subset of size 3, the number of times m was updated, and the comparison with the optimal solution.